

Narander Rubans

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Education

CSI College of Engineering, Ketti
B.E. Computer Science & Engineering

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Languages

English: Fluent | **Japanese:** JLPT N2 Attempted — Results Pending (Aug/Sept 2026) | **Tamil:** Native

Technical Skills

Generative AI: Flux 2, LoRA Fine-tuning, ComfyUI, Stable Diffusion, In-painting Pipelines

AR & Vision: Snap Camera Kit SDK, Lens Studio, OpenCV

Machine Learning: PyTorch, FAISS (Vector DB), Sentence-Transformers, Pandas, NumPy, Scikit-learn

Systems & Development: Python, JavaScript (React/Node.js), Docker, Linux (Arch), Git, Shell Scripting, Tmux

Featured Projects

Generative AI & AR Fashion Platform | *ComfyUI, Flux 2, Snap Camera Kit*

- Engineered a hybrid Virtual Try-On (VTO) system for a web application, combining a 4B rectified-flow transformer (distilled and quantized) with LoRA for garment transfer
- Trained and validated custom LoRA adapters, achieving consistent garment transfers under 8 seconds on 6GB consumer GPU VRAM with offloading
- Integrated the generative backend with a React-based frontend, optimizing the pipeline for rapid inference and seamless user interaction

Temporally Consistent Video Generation Pipeline | *ComfyUI, Wan 2.2, LTX-2.3, AnimateDiff*

- Migrated video generation workflows from AUTOMATIC1111-based localized style-transfer patching (Temporal Kit, EBSynth) to native spatial-temporal diffusion models in ComfyUI
- Built ComfyUI workflows for local models such as LTX-2.3 and Wan 2.2 to render high-fidelity video sequences, specifically engineering the pipeline to eliminate inter-frame flickering
- Maintained strict visual identity and environmental consistency across complex multi-stage renders in extended generation windows

Autonomous Visual Context Agent | *Python, Docker, Local VLM, Vector Memory, SSHFS*

- Architected a microservice-based image analytics pipeline combining kernel-level filesystem events with a locally-run vision-language model for low-overhead environmental drift monitoring and automated contextual logging
- Designed a decoupled async IPC layer over a Unix Domain Socket to isolate the filesystem watcher from the VLM inference runtime
- Built a custom vector-memory layer (MemPalace) organizing semantic tracking data into queryable categories, exposed via a natural-language CLI query interface
- Extended the pipeline to ingest live remote webcam feeds over SSHFS for real-time monitoring

Rofi YouTube Client | *Shell, mpv, yt-dlp, socat*

- Built a terminal-centric media client integrating Rofi, mpv, and yt-dlp for asynchronous streaming and playback control on Arch Linux
- Designed an IPC layer over Unix sockets using socat for real-time playback-state tracking across tools
- Implemented cookie-based authentication handling to preserve native account features while bypassing anti-bot verification

Custom High-Res Audio and Video Sync Daemon | *MPD, MPV, socat, FFmpeg, Python*

- Video toggle and auto high-res video conversion script for terminal based music player
- Wrote a custom **two-way IPC** synchronization system using socat to seamlessly link Music Player Daemon (MPD) and MPV, allowing timestamps to sync back to MPD the moment a video is toggled off.
- Mapped **custom rmpc keybinds** to control scripts for a completely uninterrupted, low-latency media playback experience within a heavily customized window manager environment.
- Built a Python fuzzy-matching algorithm to automatically tie local video files to high-fidelity FLAC audio extracted via **FFmpeg**, complete with precise, automated lyric synchronization.

Relevant Coursework

Machine Learning with Python — freeCodeCamp

Scientific Computing with Python — freeCodeCamp